ii. Gastrointestinal tract enzymes (pepsin, trypsin and lipase) are given to patients suffering from indigestion.

iii. Enzyme asparaginase is used as an anticancer drug.

iv. Enzymes are used to diagonose various diseases such as AIDS.

v. Immobilized enzymes like glucose oxidase (GOD) and peroxidase (POD) are used for the estimation of blood glucose.

Table 4.3 Some important therapeutic enzymes

Enzyme	Reaction	Therapeutic use
Asparaginase	L-Asparagine + H_2O → L-aspartate + NH_3	Leukaemia
Collagenase	Collagen hydrolysis	Skin ulcers
Glutaminase	L-Glutamine + H_2O → L-glutamate + NH_3	Leukaemia
Hyaluronidase	Hyaluronate hydrolysis	Heart attack
Lysozyme	Bacterial cell wall hydrolysis	Antibiotic
Ribonuclease	RNA hydrolysis	Antiviral
β-Lactamase	Penicillin → penicilloate	Penicillin allergy
Streptokinase	Plasminogen → plasmin	Blood clots
Trypsin	Protein hydrolysis	Inflammation
Uricase	Urate + O_2 → allantoin	Gout
Urokinase	Plasminogen → plasmin	Blood clots

Serum enzymes are used as markers to detect the cellular damage which in turn helps in the diagnosis of diseases.

Table 4.4 Serum enzymes

Serum enzyme	Disease
Amylase	Acute pancreatitis
GPT or ALT	Liver diseases (hepatitis), jaundice, cirrhosis of liver
GOT or AST	Heart attack
Alkaline phosphatase	Rickets, obstructive jaundice, bone cancer, hyperparathyroidism.
Acid phosphatase	Cancer of prostate gland

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